

CHARLESTON EXECUTIVE, SC, USA

WMO: 720606

Lat: 32.701N

Lon: 80.003W

Elev: 17

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00198

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.6	32.2	10.3	9.3	35.6	17.7	13.4	40.6	22.1	54.0	19.4	54.0	4.7	0	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	93.2	79.5	91.1	79.8	90.2	79.7	84.4	87.8	82.3	87.0	81.3	86.2	8.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.1	178.6	85.5	81.5	163.8	84.2	79.5	153.0	84.3	48.7	86.8	46.4	84.4	44.8	86.5	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.2	17.2	15.7	DB	25.4	98.0	2.8	2.5	23.4	99.8	21.7	101.3	20.1	102.7	18.0	104.6
			WB	22.3	84.1	3.3	1.0	19.9	84.9	18.0	85.5	16.1	86.1	13.7	86.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.3	49.7	53.6	58.2	66.2	74.2	80.4	82.9	81.7	78.0	69.4	58.3	54.8
	DBStd	13.35	9.50	8.64	8.46	6.36	5.14	3.92	3.05	2.83	3.82	6.63	7.92	8.67
	HDD50	275	124	58	25	0	0	0	0	0	0	0	15	53
	HDD65	1708	477	326	239	62	5	0	0	0	0	36	232	331
	CDD50	6607	115	158	280	486	750	913	1019	982	839	601	263	201
	CDD65	2564	3	7	29	98	289	463	554	517	390	171	29	14
	CDH74	24199	6	22	127	486	2088	4891	6477	5546	3344	1103	89	20
	CDH80	8517	0	0	18	57	510	1855	2725	2098	1042	208	4	0
Wind	WSAvg	6.5	6.4	7.1	7.8	8.0	7.5	6.3	6.4	5.3	5.8	6.0	6.2	5.7
Precipitation	PrecAvg	48.4	3.4	3.3	3.6	3.0	2.9	5.3	5.7	6.3	5.0	4.1	2.7	3.1
	PrecMax	62.2	8.0	10.6	6.5	6.1	6.5	10.1	8.9	11.1	10.6	12.4	7.3	9.3
	PrecMin	32.4	0.5	1.2	0.4	0.9	0.6	2.6	1.9	3.5	0.4	0.0	0.1	0.8
	PrecStd	7.4	2.0	2.1	1.6	1.5	1.7	2.2	1.9	2.2	2.4	3.1	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	77.0	81.8	84.4	92.7	96.6	96.8	95.0	91.3	88.0	79.5	76.8
		MCWB	66.3	67.5	66.3	70.0	74.2	78.4	80.9	80.3	78.2	76.9	70.5	69.8
	2%	DB	71.5	73.0	77.3	81.4	88.1	92.8	93.2	91.3	89.7	84.1	76.7	73.0
		MCWB	64.3	64.6	65.7	69.9	74.4	78.8	80.5	80.5	78.3	75.5	68.5	68.4
	5%	DB	66.3	70.0	73.3	79.3	84.2	90.5	91.2	90.2	87.9	82.0	73.4	71.7
		MCWB	61.7	63.1	63.5	68.7	73.8	79.1	80.8	80.4	77.3	73.5	67.5	67.6
	10%	DB	63.9	66.2	70.5	76.9	82.1	88.3	90.1	88.3	85.8	80.8	71.9	68.2
		MCWB	59.9	61.6	62.9	67.7	72.9	78.2	80.5	79.8	77.3	72.5	66.4	65.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.6	70.2	70.3	75.2	78.4	83.5	88.4	88.4	81.6	80.0	74.2	72.8
		MCDB	71.6	74.0	76.1	79.8	86.7	88.9	89.6	89.1	85.2	82.3	77.0	75.3
	2%	WB	65.9	67.2	68.2	73.4	76.8	82.2	85.6	84.3	80.6	78.3	71.8	70.5
		MCDB	69.3	71.2	74.4	78.2	83.4	87.3	87.9	86.4	84.8	81.9	74.5	72.0
	5%	WB	63.6	64.9	66.4	71.6	75.7	81.1	82.7	82.0	79.3	76.4	69.6	68.1
		MCDB	65.1	67.6	71.5	76.7	82.0	86.3	87.7	86.1	84.1	80.9	72.2	69.6
	10%	WB	61.5	62.8	64.7	70.0	74.6	79.9	81.5	80.9	78.3	74.6	67.1	65.6
		MCDB	63.7	65.2	68.4	74.8	80.4	85.6	86.8	85.5	83.4	78.9	70.9	68.2
Mean Daily Temperature Range	5% DB	MDBR	18.2	18.5	18.3	17.3	15.1	14.8	13.6	13.1	14.4	17.4	19.4	16.4
		MCDBR	20.1	19.7	20.0	17.9	17.5	17.2	15.4	15.6	16.8	17.3	18.8	17.6
		MCWBR	14.1	13.2	11.1	9.4	7.9	7.3	7.6	7.1	7.3	9.3	12.2	12.2
	5% WB	MCDBR	16.9	17.1	17.3	14.5	14.0	15.0	13.9	14.0	13.2	13.9	13.7	14.8
		MCWBR	14.5	13.7	11.0	8.4	6.9	7.6	8.5	7.9	6.9	8.7	10.8	12.6
Clear-Sky Solar Irradiance	taub	0.329	0.337	0.366	0.408	0.464	0.496	0.529	0.511	0.435	0.390	0.351	0.335	
	taud	2.540	2.515	2.452	2.317	2.208	2.169	2.089	2.138	2.344	2.449	2.513	2.550	
	Ebn at Noon	283	292	290	280	264	255	246	248	264	269	272	274	
	Edh at Noon	27	30	34	41	46	48	51	48	38	32	27	25	
All-Sky Solar Radiation	RadAvg	906	1119	1480	1896	2049	2072	1988	1796	1523	1317	1010	807	
	RadStd	53	115	137	128	134	143	119	116	127	139	70	64	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+360	+250

Nomenclature: See separate page